

JKBOSE 2026

CLASS 11TH

Bhaiichara
ON
TOP.

#XUV♥
HYDROCARBONS

LECTURE-04 **CHEMISTRY**

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Topics To Covered

- 1** Chemical Properties of Alkanes.

REVISE

SOURCE of alkanes.

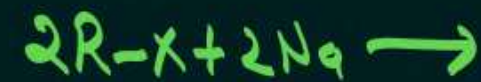
Hydrogenation



From alkyl halide

i) Red

ii) Wurtz Rxn.



From $\leftarrow \text{COOH}$.

a) Decarboxylation

b) Kolbe Electrolysis.

CHEMICAL REACTIONS OF ALKANES.

S C S C R A P

SC

SC

RAP

SUBSTITUTION
RXN.

COMBUSTION.

RXN WITH
STEAM

CONTROLLED
OXIDATION.

REARRANGEMENT
RXN
OR
ISOMERISATION.

AROMATISATION

PYROLYSIS
OR
CRACKING

Chemical Properties of Alkanes



1 Substitution Reactions

Alkanes, on reaction with halogens in presence of heat or light undergo substitution e.g.



• Reactivity order of halogens.



• Reactivity order Alkanes



Reaction with F_2 is exothermic.

Reaction with I_2 is reversible.



Because HI is strong reducing agent

That is why this particular rxn is done in presence of oxidising agent.

Mechanism $\Rightarrow$ This reaction follows free radical mechanism.

a) INITIATION $\Rightarrow$



b) PROPAGATION $\Rightarrow$



c) TERMINATION $\Rightarrow$



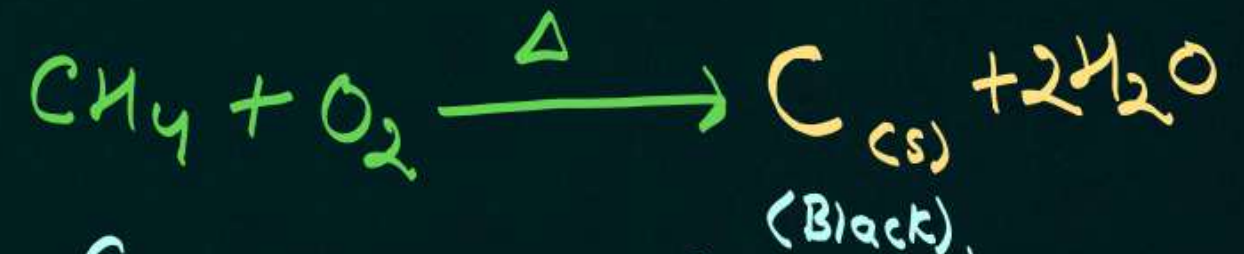
Chemical Properties of Alkanes

b Combustion

- When alkanes are heated in excess of air or dioxygen, they are oxidised to give **carbon dioxide** and **water**. A large amount of heat is evolved during this reaction.
- Thus, alkanes are used as fuels.



But if we heat alkanes with limited supply of oxygen i.e. Incomplete combustion.



Carbon black is used in ink.

Chemical Properties of Alkanes

3 Reaction with Steam

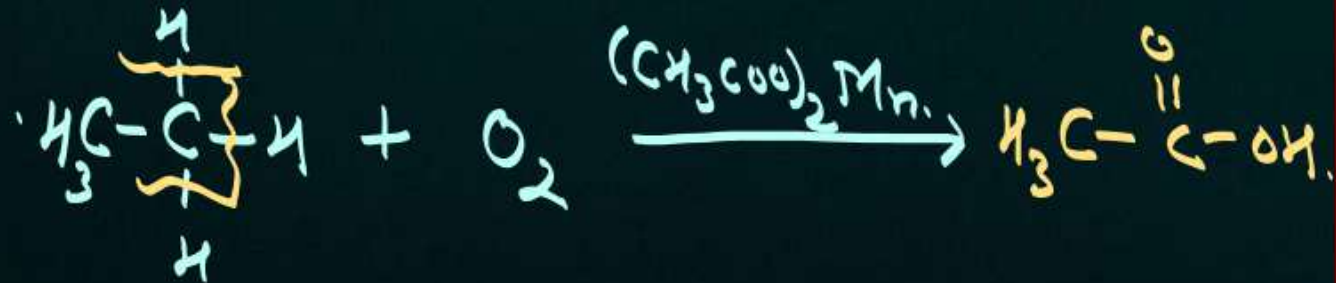
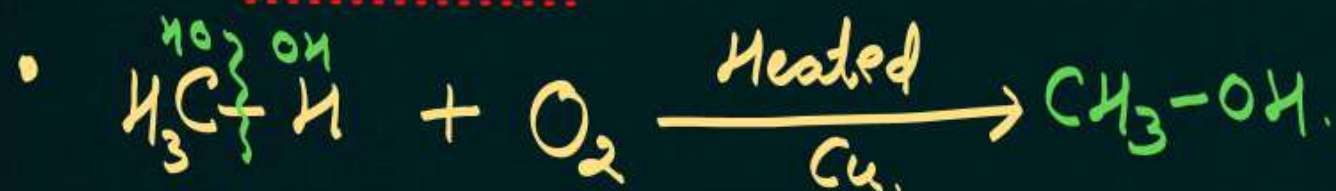
- When methane is treated with steam at 1273 K and in the presence of nickel as catalyst, it gives carbon monoxide and dihydrogen gas.



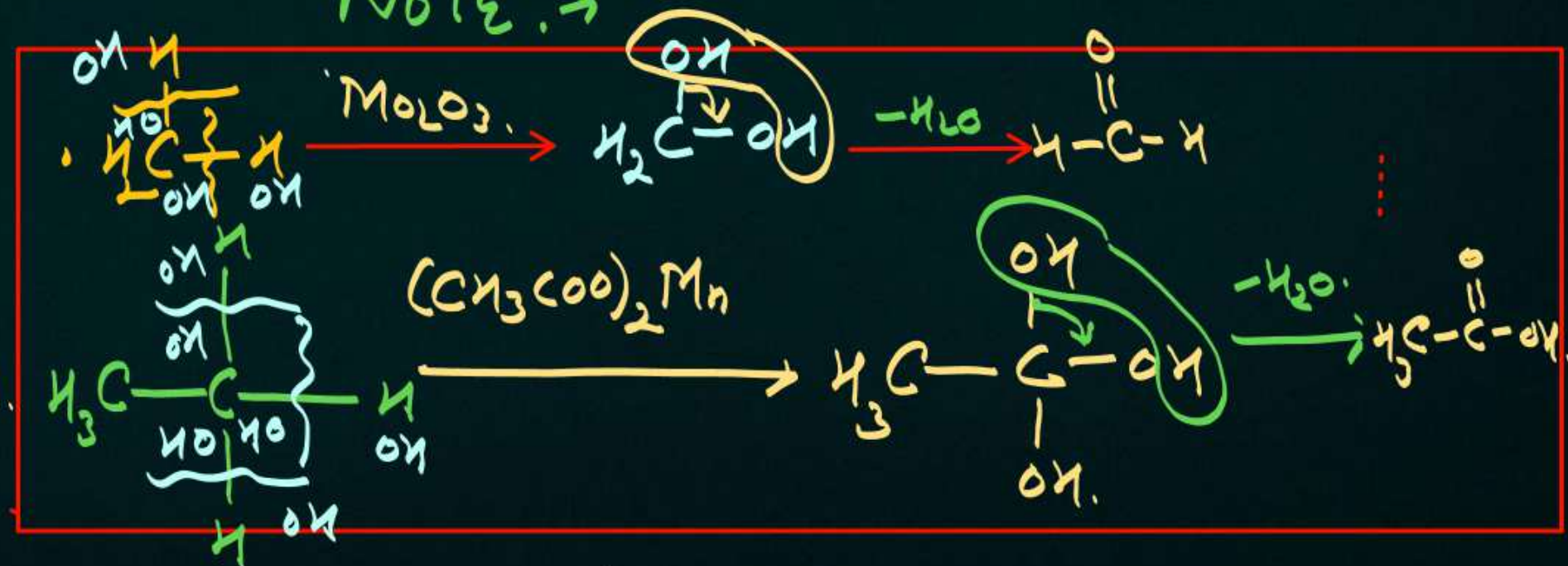
Chemical Properties of Alkanes

4 Controlled Oxidation

- If alkanes are heated in controlled supply of dioxygen or air and in the presence of suitable catalysts, a variety of oxidation products are obtained as follows



NOTE: →



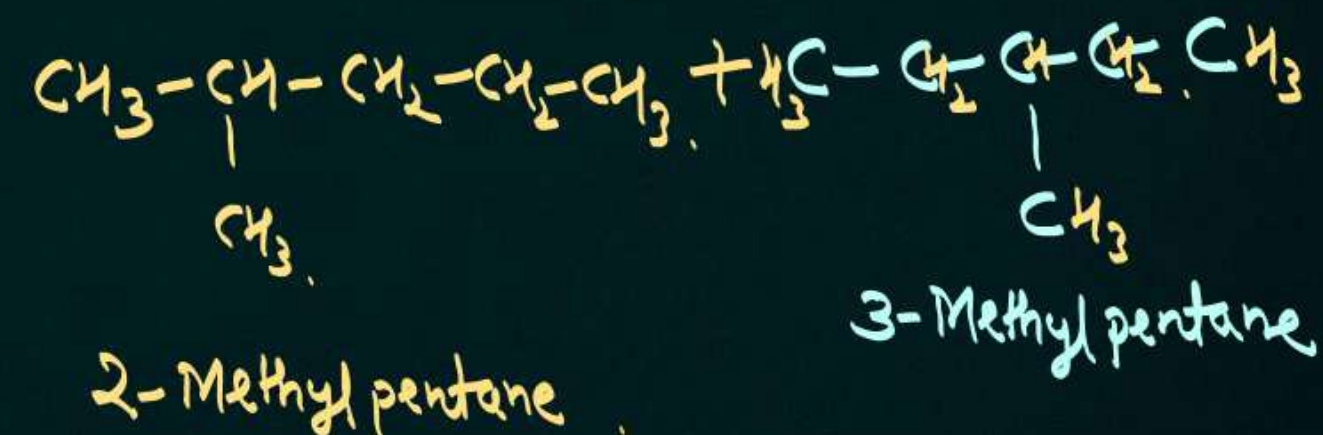
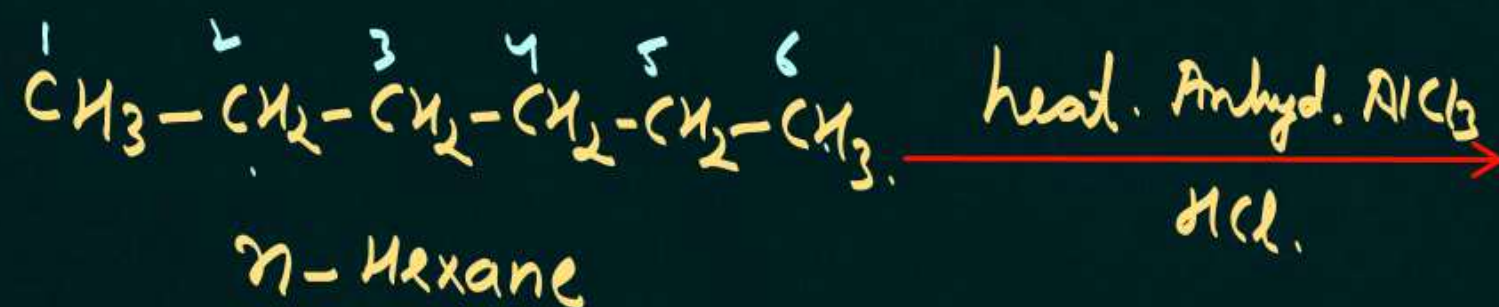
Chemical Properties of Alkanes



SLSLR (A) (P)

5 Isomerisation OR Rearrangement.

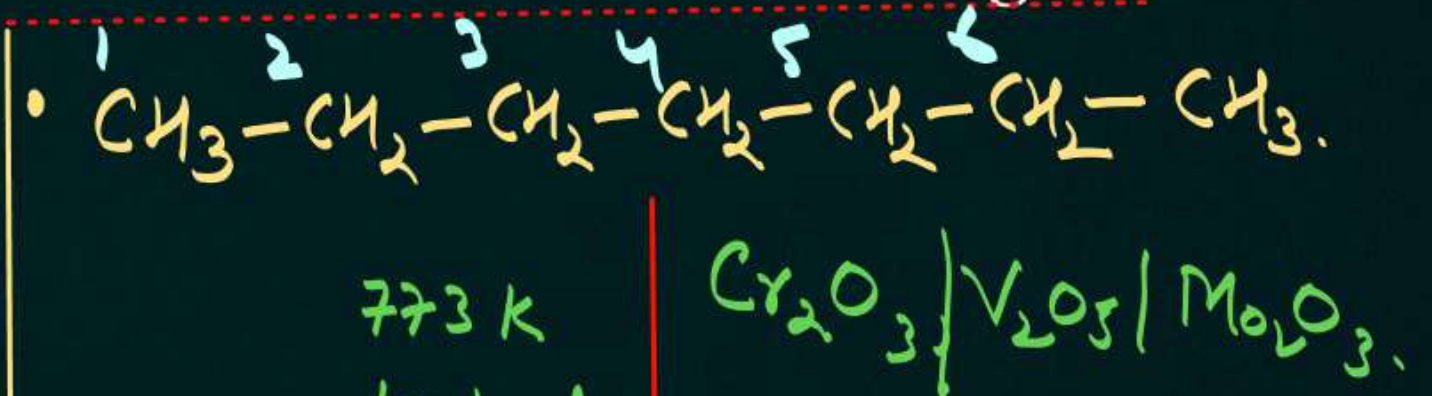
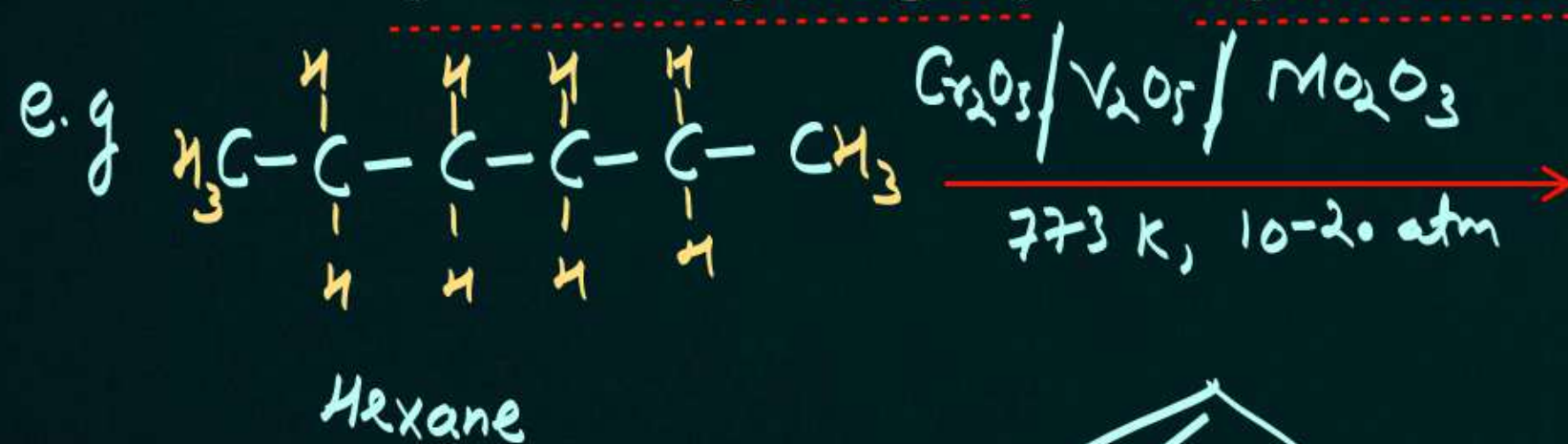
- When n -alkanes are heated with anhydrous $AlCl_3$ and HCl gas, they isomerise to give branched chain alkanes as the major product which are given below.



Chemical Properties of Alkanes

6 Aromatisation

- Straight chain alkanes having 6 or more C-atoms when heated to 773 K at 10-20 atm pressure in the presence of Cr_2O_3 , V_2O_5 or Mo_2O_3 ^{gets} dehydrogenated (i.e. loss hydrogen) and cyclised to give benzene and its homologous.

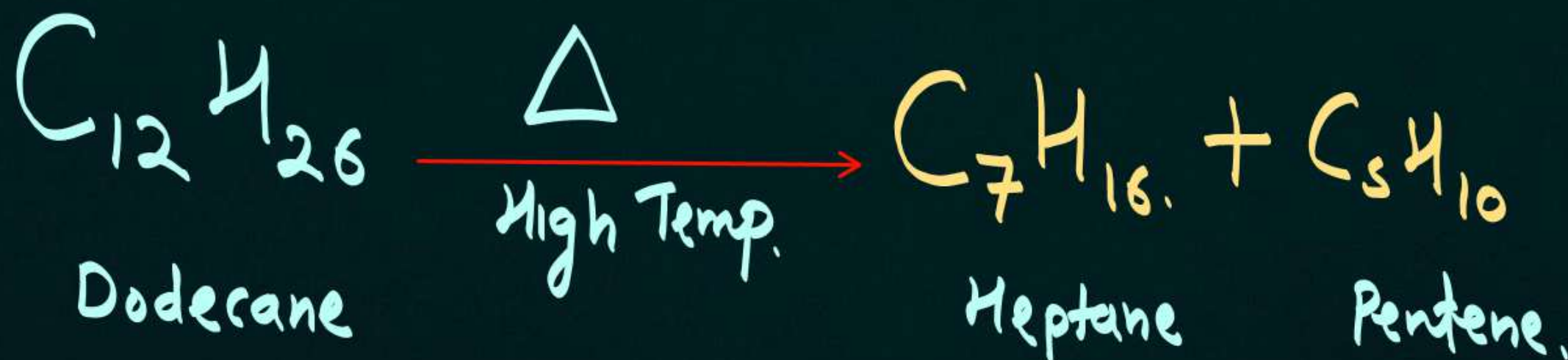


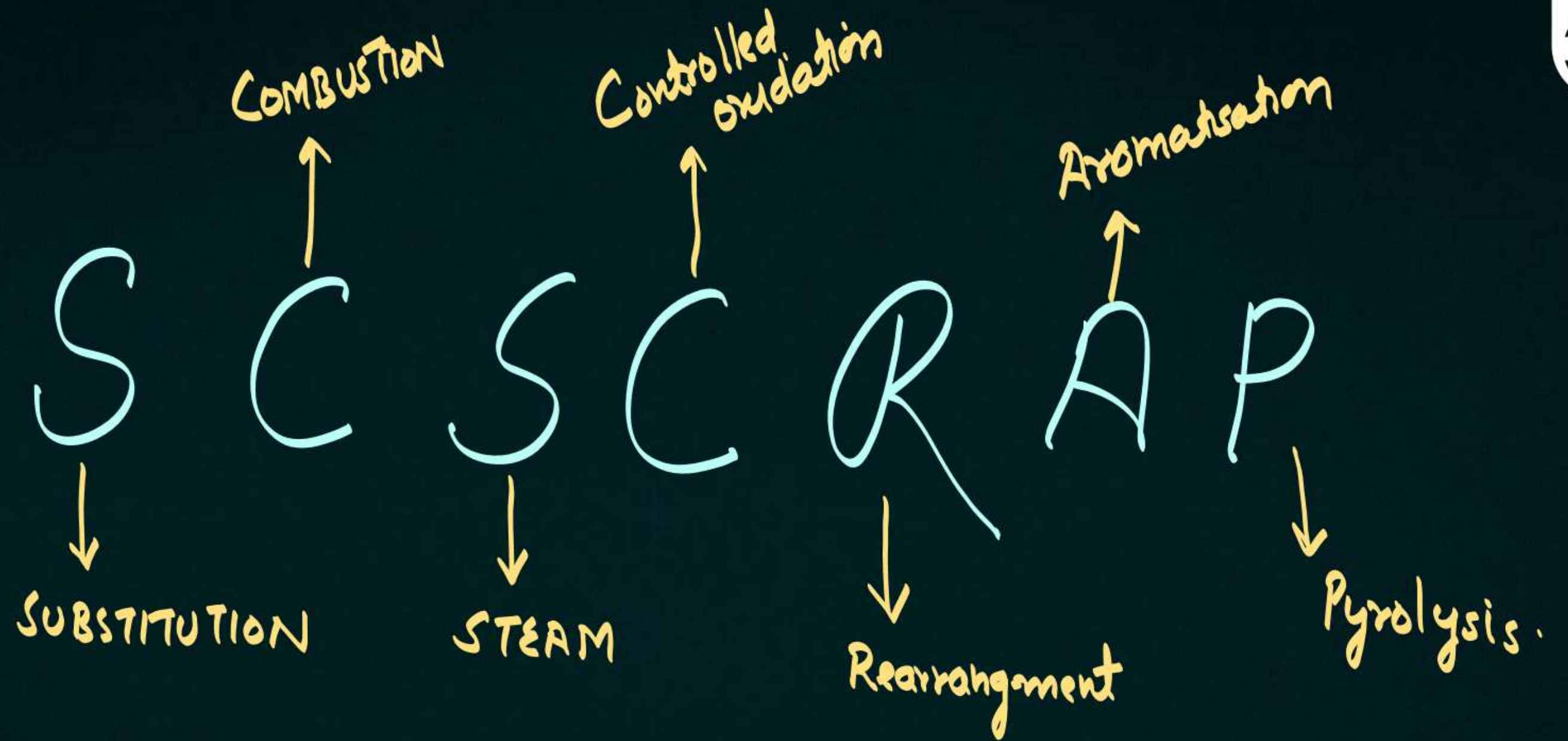
Methyl Benzene
or
Toluene.

Chemical Properties of Alkanes

7 Pyrolysis or Cracking

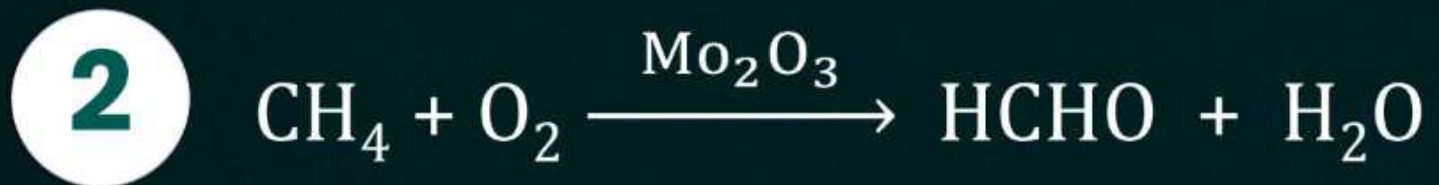
- Higher alkanes when heated at a higher temperature get decompose to give smaller fragments like lower alkanes, alkenes etc. This reaction is called pyrolysis, e.g.





QUESTION

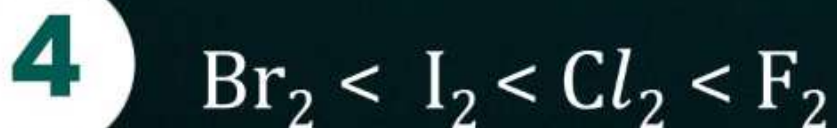
Which of the following reactions of methane is incomplete combustion?



QUESTION

Arrange the halogens F_2 , Cl_2 , Br_2 , I_2 , in order of their increasing reactivity with alkanes.

PYQ.



Bada > Chota.

- ① UZAIR SOMAIL
- ② Sehar Shah
- ③ SIMRAN DEVI

MOP
&
CHEMICAL REACTION

Likh Ke Revise KARNA.

Thank
You